

SAMBP Technical Report TR-69/2028

Type-4 Full-Converter Wind Turbine Model for SAMBP

6-State Dynamic Model: Three Turbine Presets, FRT Compliance, 150-Scenario Validation

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Contents

1	Background and Objectives	1
1.1	Motivation	1
1.2	Objectives	1
2	Theory and Methodology	1
2.1	6-State Model	1
2.2	Turbine Presets	2
2.3	FRT Compliance	2
3	Simulation Results	2
3.1	150-Scenario Summary	2
3.2	GFM vs GFL Fault Current Comparison	2
3.3	Key Findings	3
4	Conclusion	3

1 Background and Objectives

1.1 Motivation

Type-4 (full-converter) wind turbines decouple the generator from the grid through a back-to-back voltage source converter (VSC). This decoupling eliminates the stator transient DC offset characteristic of Type-3 (DFIG) machines and means the fault current is entirely determined by the grid-side converter (GSC) current control [1]. Modern Type-4 turbines range from 8 MW (Vestas V164) to 16 MW (Goldwind GWH252), and grid-forming (GFM) variants (e.g., Siemens Gamesa SG14 with VSM control) are entering commercial deployment.

The SAMBP framework previously modelled Type-3 DFIG (TR-56) and PV (TR-62) IBRs. TR-69 extends coverage to Type-4 turbines, which dominate offshore wind procurements. The 6-state model presented here captures aerodynamic, mechanical, and electrical dynamics at a fidelity suitable for protection studies (millisecond timescale) while remaining computationally tractable for 150-scenario sweeps.

1.2 Objectives

1. Develop a 6-state dynamic model covering $P_{\text{gen}}, Q_{\text{gen}}, V_{\text{dc}}, I_{\text{grid}}, \omega_{\text{rotor}}, \beta$ (pitch angle).
2. Implement three turbine presets: Vestas V164-8MW (GFL), Siemens SG14-222 (GFM/VSM), Goldwind GWH252-16MW (GFL).
3. Validate FRT compliance against IEC 61400-27-1, AEMO NER S5.2.5.13, and NERC PRC-024-2.
4. Confirm 100% scenario agreement over 150 scenarios.

2 Theory and Methodology

2.1 6-State Model

The state vector is $\mathbf{x} = [P_{\text{gen}}, Q_{\text{gen}}, V_{\text{dc}}, I_{\text{grid}}, \omega_{\text{rotor}}, \beta]^T$.

Aerodynamic power. The mechanical power extracted from wind is:

$$P_m = \frac{1}{2} \rho A C_p(\lambda, \beta) v_w^3 \quad (1)$$

where $C_p(\lambda, \beta)$ is the 6-parameter Heier polynomial [2]:

$$C_p = c_1 \left(\frac{c_2}{\lambda_i} - c_3 \beta - c_4 \right) e^{-c_5/\lambda_i} + c_6 \lambda, \quad \frac{1}{\lambda_i} = \frac{1}{\lambda + 0.08\beta} - \frac{0.035}{\beta^3 + 1} \quad (2)$$

with $(c_1, \dots, c_6) = (0.5176, 116, 0.4, 5, 21, 0.0068)$.

MPPT control. Maximum power point tracking sets the rotor speed reference: $\omega_{\text{ref}} = k_{\text{opt}} v_w$, yielding $P_{\text{mppt}} = k_{\text{opt}}^3 v_w^3 / (2\rho A)$.

Pitch PI controller. Above rated wind speed, pitch angle β is regulated by: $\dot{\beta} = K_p^\beta (\omega - \omega_{\text{rated}}) + K_i^\beta \int (\omega - \omega_{\text{rated}}) dt$, with $\beta_{\text{min}} = 0^\circ$ and $\beta_{\text{max}} = 90^\circ$.

DC link dynamics. $C_{\text{dc}} \dot{V}_{\text{dc}} = (P_m - P_{\text{gen}}) / V_{\text{dc}}$.

Grid-side converter. For GFL mode: I_{grid} tracks a current reference from the power setpoints via a PI current controller in dq frame. For GFM (VSM) mode (SG14 preset): V_{grid} is imposed by the VSM swing equation with virtual inertia $H_v = 5$ s and droop $D_v = 0.05$ pu.

2.2 Turbine Presets

Table 1: Type-4 turbine preset parameters.

Parameter	Vestas V164-8MW	Siemens SG14 (GFM)	Goldwind GWH252-16MW
Rated power (MW)	8.0	14.0	16.0
Rotor diameter (m)	164	222	252
Rated wind speed (m/s)	12.0	10.5	11.5
Control mode	GFL	GFM/VSM	GFL
Virtual inertia H_v (s)	—	5.0	—
Droop D_v (pu)	—	0.05	—
I_{max} (pu)	1.10	1.10	1.10

2.3 FRT Compliance

FRT capability is validated against three grid codes:

IEC 61400-27-1: Voltage dip to 0 pu for 150 ms; reactive current injection $\Delta I_q = k_{\text{ftr}} \Delta V$, $k_{\text{ftr}} = 2.0$.

AEMO NER S5.2.5.13: Voltage dip to 0.15 pu for 200 ms; active power recovery to 90% within 1 s.

NERC PRC-024-2: Voltage ride-through envelope: no trip for $V \geq 0.15$ pu up to 2 s.

3 Simulation Results

3.1 150-Scenario Summary

Table 2: TR-69 150-scenario agreement summary.

Test Group	Scenarios	AGREE	DISAGREE	Agree Rate	Status
T1: GFL steady-state	30	30	0	100%	✓
T2: GFL FRT (IEC 61400-27-1)	20	20	0	100%	✓
T3: GFM/VSM dynamics	30	30	0	100%	✓
T4: MPPT tracking	20	20	0	100%	✓
T5: Pitch control	20	20	0	100%	✓
T6: Protection (76AC, 87L)	20	20	0	100%	✓
T7: Multi-turbine interaction	10	10	0	100%	✓
Total	150	150	0	100%	PASS

3.2 GFM vs GFL Fault Current Comparison

During a 3-phase fault at the collector bus:

- **GFL (Vestas/Goldwind):** $I_{\text{fault}} = 1.10$ pu (hard current limit); no DC offset; angle controlled to inject maximum reactive current per IEC 61400-27-1.
- **GFM (SG14 VSM):** I_{fault} initially follows VSM swing equation (transient up to 1.5 pu for first 20 ms), then limited to 1.10 pu by inner current limiter; virtual frequency drop $\Delta f = D_v^{-1} \Delta P$ detectable by 81-element.

Table 3: Fault current characteristics: Type-4 turbine variants.

Metric	GFL (V164)	GFM/VSM (SG14)	GFL (GWH252)
Steady-state fault current (pu)	1.10	1.10	1.10
Peak transient (pu, 0–20 ms)	1.10	1.48	1.10
DC offset	None	Small	None
Reactive current injection k_{ftr}	2.0	2.0	2.0
Active power recovery (90%, s)	0.8	0.6	0.9

3.3 Key Findings

Finding 1 (GFM transient matters for protection). *The GFM/VSM variant (SG14) produces a transient fault current up to 1.48 pu in the first 20 ms before the inner current limiter engages.*

This transient is sufficient to operate a 76AC element set at 1.3 pu during an internal fault, resolving the “GFM blind spot” documented in TR-50.

Finding 2 (Pitch controller does not affect protection timescale). *The pitch controller time constant ($\tau_\beta \approx 200$ ms) is well above the protection decision window (≤ 40 ms). The 6-state model correctly decouples pitch dynamics from electrical fault response.*

4 Conclusion

TR-69 delivers the Type-4 full-converter wind turbine model for the SAMBP Digital Twin. The 6-state model with Heier C_p polynomial, MPPT, pitch PI, and VSM option is validated across 150 scenarios with 100% agreement. Three turbine presets (Vestas V164-8MW, Siemens SG14 GFM, Goldwind GWH252-16MW) are confirmed FRT-compliant under IEC 61400-27-1, AEMO NER, and NERC PRC-024-2.

References

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- [2] S. Heier, *Grid Integration of Wind Energy: Onshore and Offshore Conversion Systems*, 3rd ed., Wiley, Chichester, 2014.
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